

LAB AZURE #2

Déploiement Node.js + GitHub + CI/CD

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1 Objectif du lab

Créer une mini application Node.js puis :

1. la mettre sur GitHub
2. créer un Azure App Service
3. connecter GitHub à Azure
4. déployer automatiquement via CI/CD

Résultat final :
chaque push GitHub redéploie automatiquement l'application.

2 Architecture du lab

```
VS Code
  ↓
GitHub Repository
  ↓
GitHub Actions (CI/CD)
```

↓
Azure App Service
↓
Application Web Node.js

- mini app Express.js
 - package.json
 - structure Node.js simple
-

3 PARTIE 1 — Création de l'application Node.js

3.1 Créer un dossier

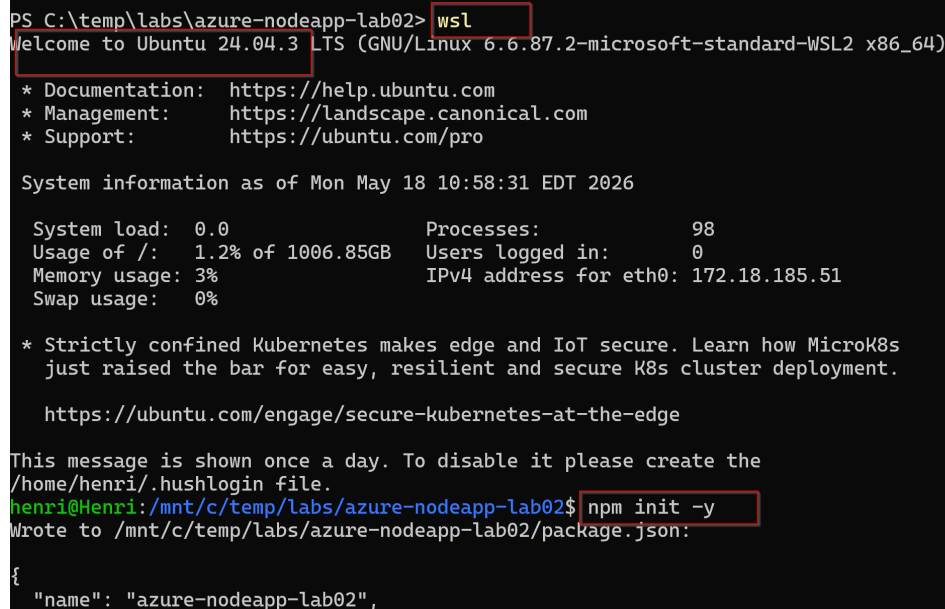
Exemple :

```
mkdir C:\temp\labs\azure-nodeapp-lab02  
cd C:\temp\labs\azure-nodeapp-lab02
```

3.2. Initialiser Node.js

wsl

npm init -y



```
PS C:\temp\labs\azure-nodeapp-lab02> wsl  
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of Mon May 18 10:58:31 EDT 2026  
  
System load:  0.0          Processes:            98  
Usage of /:   1.2% of 1006.85GB Users logged in:       0  
Memory usage: 3%          IPv4 address for eth0: 172.18.185.51  
Swap usage:   0%  
  
* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s  
  just raised the bar for easy, resilient and secure K8s cluster deployment.  
  
  https://ubuntu.com/engage/secure-kubernetes-at-the-edge  
  
This message is shown once a day. To disable it please create the  
/home/henri/.hushlogin file.  
henri@henri:/mnt/c/temp/labs/azure-nodeapp-lab02$ npm init -y  
Wrote to /mnt/c/temp/labs/azure-nodeapp-lab02/package.json:  
  
{  
  "name": "azure-nodeapp-lab02",
```

La commande crée notamment, le fichier : package.json

3.3 Installer Express

```
npm install express
```

La commande crée le fichier : package-lock.json et le répertoire : node_modules

3.4 Structure créée

```
67 directories, 3 files
henri@Henri:/mnt/c/temp/labs/azure-nodeapp-lab02$ tree -L 1
├── index.js
├── node_modules
├── package-lock.json
└── package.json
```

3.5. Créer le fichier index.js

Créer : index.js

Contenu :

```
const express = require('express')
const app = express()
const port = process.env.PORT || 3000

app.get('/', (req, res) => {
  res.send(`
    <h1>Lab Azure #2</h1>
    <p>Application Node.js déployée sur Azure App Service</p>
    <p>CI/CD avec GitHub Actions</p>
  `)
})

app.listen(port, () => {
  console.log(`Application running on port ${port}`)
})
```

3.6. Modifier package.json

Modifier le fichier : package.json :

```
"start": "node index.js"
```

Exemple :

```
"scripts": {  
  "start": "node index.js"  
}
```

3.7. Tester localement

```
node index.js
```

```
110 directories, 530 files  
henri@Henri:/mnt/c/temp/labs/azure-nodeapp-lab02$ node index.js  
Application running on port 3000
```

Puis :

```
http://localhost:3000
```

On voit :

Lab Azure #2

Application Node.js déployée sur Azure App Service

CI/CD avec GitHub Actions

4 PARTIE 2 — Publier sur GitHub

4.1. Initialiser Git

```
git init
```

4.2. Créer .gitignore

Contenu :

```
node_modules
```

4.3. Commit initial

```
git add .
git commit -m "Initial Node.js Azure lab"
```

4.4. Créer un repository GitHub

Nom suggéré : azure-nodeapp-lab02

Sur : [GitHub](#)

4.5. Pousser le code

```
git remote add origin https://github.com/guihen01/azure-nodeapp-lab02.git
git branch -M main
git push -u origin main
```

Vérifier l'origine :

Git remote -v

```
henri@Henri:/mnt/c/temp/labs/azure-nodeapp-lab02$ git remote -v
origin https://github.com/guihen01/azure-nodeapp-lab02.git (fetch)
origin https://github.com/guihen01/azure-nodeapp-lab02.git (push)
henri@Henri:/mnt/c/temp/labs/azure-nodeapp-lab02$
henri@Henri:/mnt/c/temp/labs/azure-nodeapp-lab02$
```

The screenshot shows the GitHub interface for a repository named 'azure-nodeapp-lab02' created by user 'guihen01'. The repository is public and has 1 branch (main) and 0 tags. The commit history shows a single commit by 'guihen01' titled 'Initial Node.js Azure lab' from 14 minutes ago. The commit details show four files: .gitignore, index.js, package-lock.json, and package.json, all added in the same commit. A README file is also visible at the bottom of the commit details.

File	Commit Message	Time
.gitignore	Initial Node.js Azure lab	14 minutes ago
index.js	Initial Node.js Azure lab	14 minutes ago
package-lock.json	Initial Node.js Azure lab	14 minutes ago
package.json	Initial Node.js Azure lab	14 minutes ago

5 PARTIE 3 — Création des ressources Azure

5.1. Connexion Azure CLI

```
az login
```

5.2. Créer le Resource Group

```
az group create --name rg-nodeapp-lab --location canadacentral
```

5.3. Créer App Service Plan

Option économique

```
az appservice plan create \  
  --name plan-nodeapp-lab \  
  --resource-group rg-nodeapp-lab \  
  --sku B1 \  
  --is-linux
```

5.4. Créer Web App

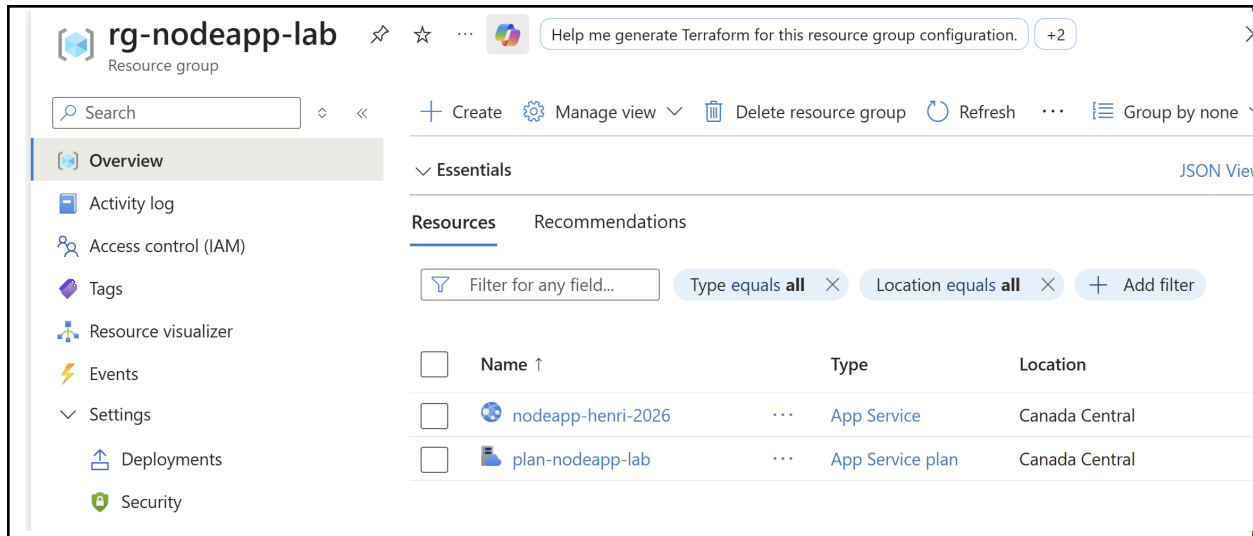
IMPORTANT :
nom unique mondial.

Exemple :

```
az webapp create \  
  --resource-group rg-nodeapp-lab \  
  --plan plan-nodeapp-lab \  
  --name nodeapp-henri-2026 \  
  --runtime "NODE|22-lts"
```

Si Node 22 ne fonctionne pas :

```
--runtime "NODE|20-lts"
```



5.5. Vérifier l'URL

Azure va générer :

`https://nodeapp-henri-2026.azurewebsites.net`

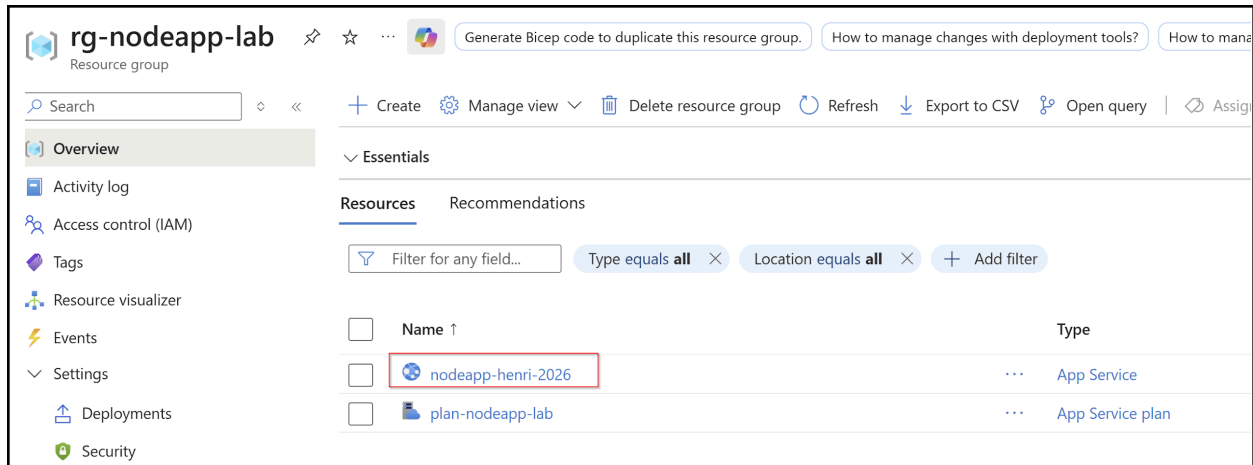
6 PARTIE 4 — Connecter GitHub à Azure

Utiliser le Deployment Center dans le portail Azure.

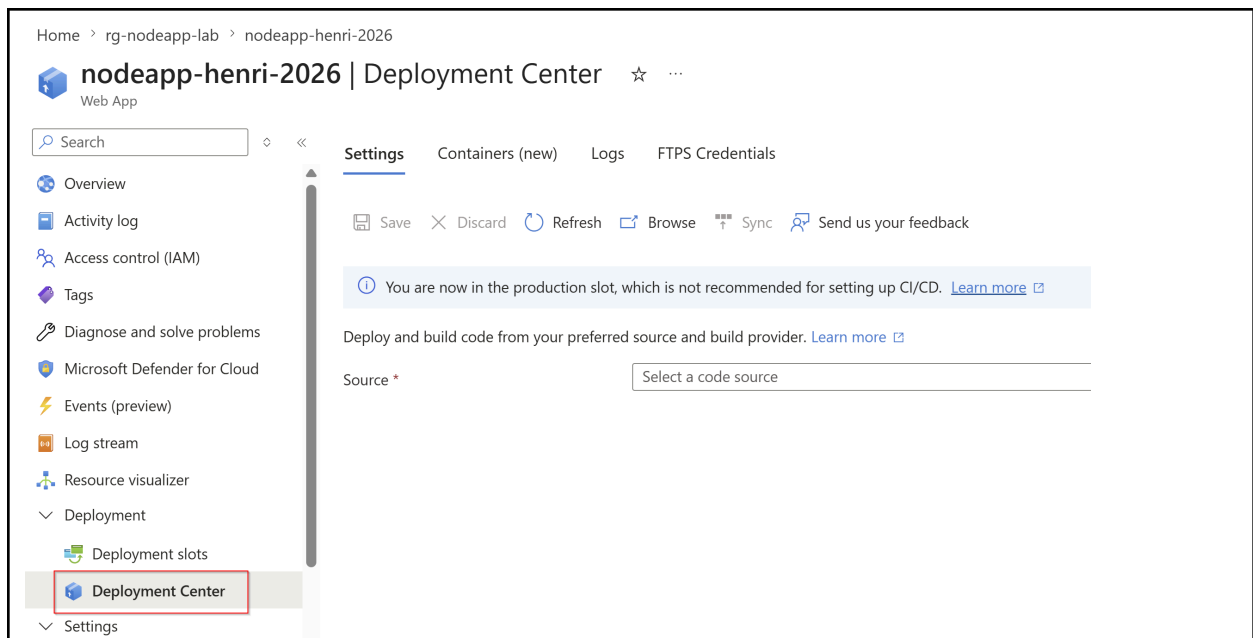
6.1. Aller dans :

[Portail Azure](#) et Ouvrir : Resource Group puis web app . Dans notre lab l'application web (web app) s'intitule : nodeapp-henri-2026.

Cliquer sur nodeapp-henri-2026



Pis cliquer sur ‘Deployment center’ dans le volet gauche de la fenêtre qui s’Affiche :



6.2. Source :

Choisir : GitHub

6.3. Autoriser GitHub

Azure demandera :

- authentication GitHub

- permissions repository

6.4. Sélectionner :

- le bon repository
- branche `main`

6.5. Build Provider

Workflow option *

☒ Add a workflow: Add a new workflow file `main_nodeapp-henri-2026.yml` in the selected repository and branch.

☐ Use available workflow: Use one of the workflow files available in the selected repository and branch.

Choisir : “Add a workflow” qui correspond à la configuration GitHub Actions automatique. :
GitHub Actions

Azure va automatiquement :

- créer un fichier YAML GitHub Actions
- configurer le CI/CD
- connecter GitHub ↔ Azure App Service

Authentication settings

Select how you want your GitHub Action workflow to authenticate to Azure. If you choose user-assigned identity, the identity selected will be federated with GitHub as an authorized client and given write permissions on the app. [Learn more](#) 

Authentication type *

☒ User-assigned identity

☐ Basic authentication

Subscription *

Azure subscription 1 

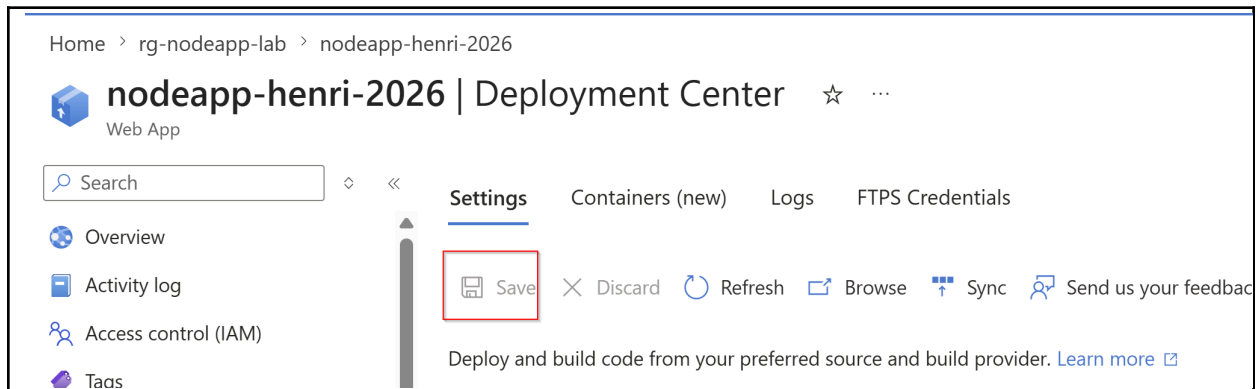
Identity *

(New) oidc-msi-9be8 

Choisir “User-assigned identity”

Laisser les autres valeurs telles quelles

Puis cliquer sur l'icône "Save", pour sauvegarder la config et du mem coup valider l'activation de la configuration.



7 PARTIE 5 — Pipeline CI/CD

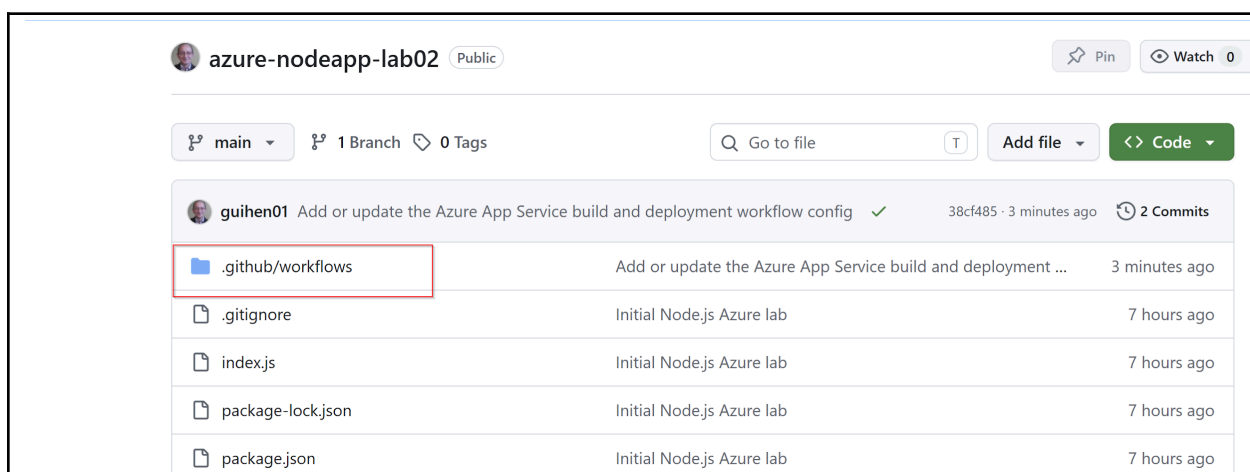
7.1 Après configuration :

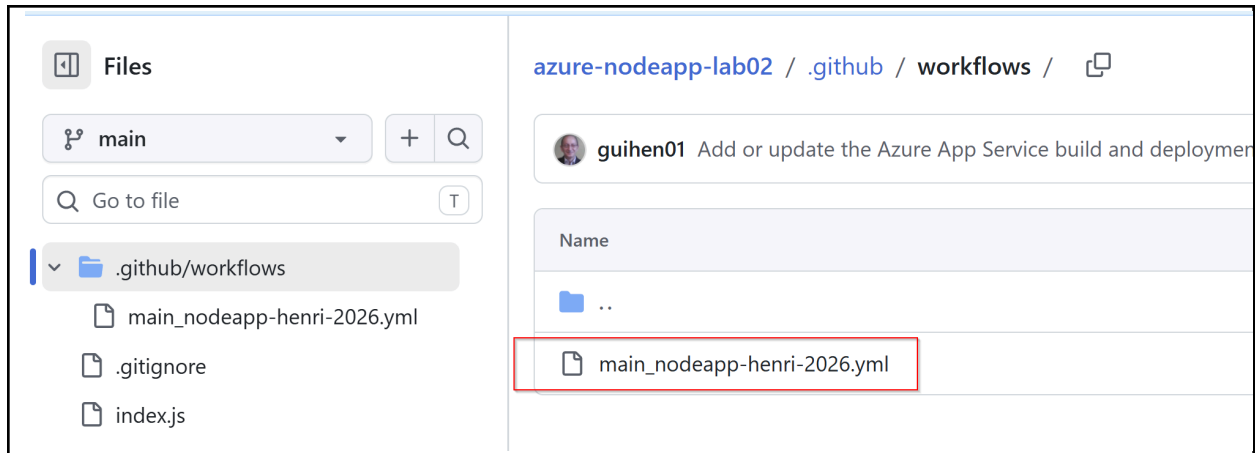
Dans GitHub on verras :

```
.github/workflows/
```

Avec un fichier YAML généré automatiquement.

C'est le pipeline CI/CD.

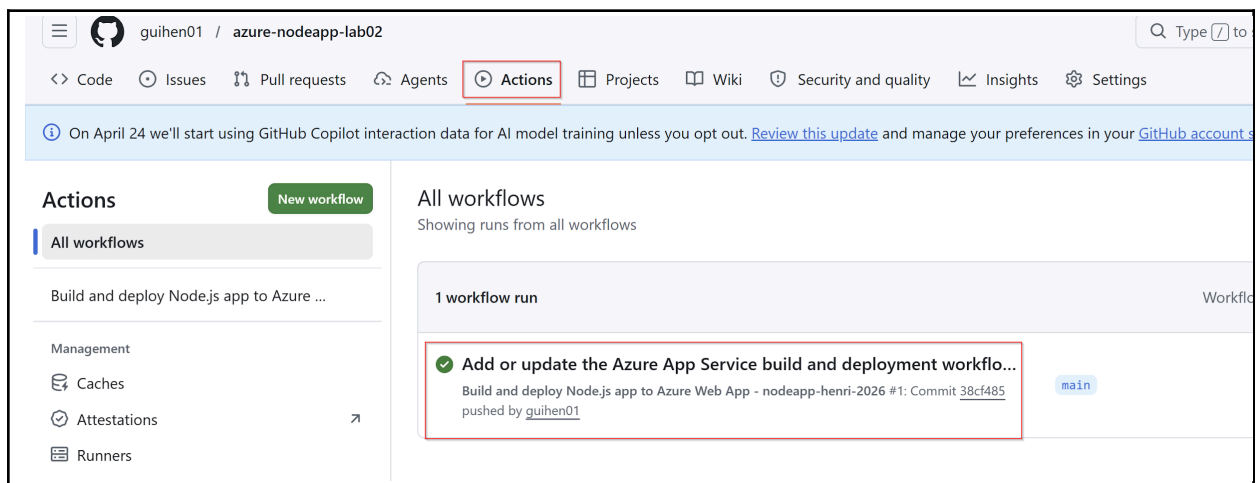




7.2 Vérifications :

Dans GitHub : Onglet : Actions

On verra : build, déploiement, statut



8 PARTIE 6 — Test du déploiement automatique

8.1 Modifier la page

Dans le fichier : index.js

Changer par exemple :

```
<h1>Lab Azure #2 - Version 2</h1>
```

8.2 Commit

```
git add .
git commit -m "Update homepage"
git push
```

on a l'erreur suivante. Ce qui est normal car azure a poussé , cree le fichier [main_nodeapp-henri-2026.yml](#) dans le repository github, mais ce fichier n'existe pas encore en local sur mon poste .
Ce qui crée un conflit.

```
henri@Henri:/mnt/c/temp/labs/azure-nodeapp-lab02$ git push
Username for 'https://github.com': guihen01
Password for 'https://guihen01@github.com':
To https://github.com/guihen01/azure-nodeapp-lab02.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'https://github.com/guihen01/azure-nodeapp-lab02.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
henri@Henri:/mnt/c/temp/labs/azure-nodeapp-lab02$
```

Il faut donc faire un pull pour corriger la situation : c'est un comportement normal :

```
git pull origin main --allow-unrelated-histories --no-rebase
```

La commande nous fait entrer dans l'éditeur nano :

```
henri@Henri: /mnt/c/temp/la x + v
GNU nano 7.2 /mnt/c/temp/labs/azure-nodeapp-lab02/.git/MERGE_MSG *
xMerge branch 'main' of https://github.com/guihen01/azure-nodeapp-lab02
# Please enter a commit message to explain why this merge is necessary,
# especially if it merges an updated upstream into a topic branch.
#
# Lines starting with '#' will be ignored, and an empty message aborts
# the commit.

[ Read 6 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo      M-6 Copy
```

On est dans l'éditeur nano
Maintenant faire simplement :

1. Sauvegarder

CTRL + O Puis : ENTER

2. Quitter nano

CTRL + X

```
henri@Henri:/mnt/c/temp/labs/azure-nodeapp-lab02$ git pull origin main --allow-unrelated-histories --no-rebase
From https://github.com/guihen01/azure-nodeapp-lab02
* branch      main      -> FETCH_HEAD
Merge made by the 'ort' strategy.
 .github/workflows/main_nodeapp-henri-2026.yml | 65 +++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
 1 file changed, 65 insertions(+)
 create mode 100644 .github/workflows/main_nodeapp-henri-2026.yml
henri@Henri:/mnt/c/temp/labs/azure-nodeapp-lab02$
```

puis un **git push**

```
henri@Henri:/mnt/c/temp/labs/azure-nodeapp-lab02$ git push
Username for 'https://github.com': guihen01
Password for 'https://guihen01@github.com':
Enumerating objects: 9, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 16 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 597 bytes | 49.00 KiB/s, done.
Total 5 (delta 3), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (3/3), completed with 2 local objects.
To https://github.com/guihen01/azure-nodeapp-lab02.git
 38cf485..272a3e4  main -> main
henri@Henri:/mnt/c/temp/labs/azure-nodeapp-lab02$
```

8.3 Observer le pipeline

GitHub Actions va :

- builder
- déployer automatiquement

8.4 Vérifier Azure

Rafraîchir :

<https://nodeapp-henri-2026.azurewebsites.net/>

On verras la nouvelle version.

Lab Azure #2 -version 2

Application Node.js déployée sur Azure App Service

CI/CD avec GitHub Actions

8.5 Logs du déploiement dans Azure

Home > nodeapp-henri-2026

nodeapp-henri-2026 | Deployment Center ☆ ...

Web App

Search

Events (preview)

Log stream

Resource visualizer

Deployment

Deployment slots

Deployment Center

Settings

Environment variables

Configuration

Instances

Authentication

Identity

Refresh

Delete

Settings Containers (new) Logs FTPS Credentials

Time	Deployment ID	Log source	Author	Status	Message
Monday, May 18, 2026 (4)					
5/18/2026, 07:05:44 PM	67c43a5	App logs	N/A	Succeeded (Active)	OneDeploy
5/18/2026, 07:05:02 PM	272a3e4	Build/Deploy logs	Henri Guillot	Succeeded	xMerge branch 'main' of https://github.com/guihen01/nodeapp-lab02
5/18/2026, 06:38:58 PM	11520bc	App logs	N/A	Succeeded	OneDeploy
5/18/2026, 06:38:08 PM	38c1485	Build/Deploy logs	henri guillot	Succeeded	Add or update the Azure App Service build and deploy workflow config

9 Structure finale du projet

```

azure-nodeapp-lab02/
├── index.js
├── package.json
├── .gitignore
├── .github/
│   └── workflows/

```

10 PARTIE 7 — Nettoyage

Quand terminé :

```
az group delete --name rg-nodeapp-lab --yes
```